

Yoosung Hong

Game Technology Software Engineer

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Game technology software engineer with 1+ years of industry experience building Unreal Engine 5 VR and simulation systems under real-time and hardware constraints. Delivered on-device AI and voice pipelines, reduced end-to-end latency by more than 50%, and ran UE5, a local LLM, and text-to-speech workloads concurrently within an 8 GB VRAM budget. Contributor to AMD Schola with a merged fix for RLLib-Schola agent synchronization. Strong in C++, asynchronous systems, SIMD signal processing, and data-driven experimentation across production engineering, open source, and game AI research.

SKILLS

- Languages: C++, C#, Python
- Game & Real-Time Systems: Unreal Engine 5, Unity, VR
- AI & Simulation: PyTorch, reinforcement learning, LLM fine-tuning, llama.cpp, LangGraph, Ollama
- Systems & Performance: Asynchronous task design, SIMD, on-device inference, real-time audio, latency and memory optimization, Linux

WORK EXPERIENCE

SamwoolImmersion - VR & AI Engineer

Mar 2025 - Jul 2026

- Developed and integrated Unreal Engine 5 AI features for interactive VR and simulation products, owning work from technical PoC and system design through deployment and maintenance.
- Built on-device inference and real-time voice interaction pipelines for constrained hardware, integrating components across the game engine, model, and service layers.
- Improved runtime responsiveness through model optimization, asynchronous processing, and low-level audio data handling.
- Conducted applied R&D and converted results into production features, technical documentation, a patent filing, and internal technology transfer.
- Deployed supporting services with Docker, Terraform, and AWS, and built internal automation agents for operational workflows.

OPEN SOURCE HIGHLIGHT

AMD Schola - Contributor

Merged Pull Request

- Diagnosed and fixed an RLLib-Schola synchronization defect affecting terminated agents; the fix was reviewed and merged into AMD GPUOpen's official repository.

SELECTED ENGINEERING PROJECTS

TSR - Unreal Engine 5 VR AI Tutoring System

Mar 2025 - Nov 2025

Lead Programmer | SamwooImmersion

- Led development of a UE5 VR application that ran rendering, a local LLM, and text-to-speech concurrently within an 8 GB VRAM budget through rendering and model optimization.
- Fine-tuned Gemma 3 1B with LoRA to distill persona prompts, reducing prompt size to 20% of the original and end-to-end response time by 33%.
- Deployed supporting web services using Docker, Terraform, and AWS.

VR Voice Interactive Plugin

Nov 2025 - Feb 2026

Lead Programmer | SamwooImmersion

- Developed a standalone UE5 VR HMD plugin for real-time spoken keyword detection using Sherpa-ONNX Runtime, achieving sub-200 ms recognition latency.
- Implemented a custom audio resampler with ring buffers and SIMD to bridge device and model input formats.
- Separated resampling and inference into asynchronous tasks, reducing end-to-end latency by more than 50%.

Dynamic EQS

Aug 2025 - Mar 2026

Lead Programmer | Personal Project

- Developed a UE5 plugin that applies reinforcement learning to optimize Environment Query System weights in a multi-agent combat simulation built with the Gameplay Ability System.
- Built a distributed MAPPO training pipeline with Ray RLlib and AWS; an attention-based spatial representation improved win rate by 2.3x and distributed object placement by 43%.
- Converted the engineering results into a sole-author research paper accepted as a poster at the ICML 2026 NExT-Game Workshop.

Virtual Factory Simulation using AI agents

Mar 2026 - Present

Lead Programmer | Personal Project

- Developed an Unreal Engine 5 Pixel Streaming platform for browser-based control of a virtual process simulation.
- Implemented agent planning, tool routing, closed-loop execution, and a validation layer using LangGraph and GraphRAG.
- Fine-tuned Qwen3.5-2B with supervised learning, improving tool-routing success from 49% to 96%, and built a llama.cpp runtime with switchable LoRA adapters.

SELECTED RESEARCH

Learned Coordination Conventions in Cooperative MARL

ICML 2026 Workshop

- Poster accepted at NExT-Game 2026: New Frontiers in Game-Theoretic Learning; sole author.

One Policy, Infinite NPCs

arXiv Preprint

- Scalable persona-conditioned NPC control through shared reinforcement learning policies; sole author.

GOBT: A Synergistic Approach to Game AI

JMIS, Vol. 10, No. 4, 2023

- First-author journal article on hybrid planning and decision-making for game agents.

EDUCATION

Dong-eui University

Mar 2019 - Feb 2025

· B.S. in Game Engineering

Deep Learning Specialization

Feb 2026 - Jun 2026

· Deep Neural Networks, Convolutional Neural Networks, Sequence Models

PATENTS

Korean Patent Application 10-2025-012****

First Inventor

· Training method and system for teacher corrective guidance using generative-AI-based question-type classification.

AWARDS

- Best Paper Award, Korean Multimedia Society Spring Conference, 2024.
- Excellence Award and Bronze Award, Korea Digital Contents Society Fall Conference, 2023.
- 10 university project and capstone competition awards.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (Jan 2026)
- Information Processing Engineer, Korea (Sep 2025)
- Linux Master Level 2 (Oct 2025)